Experiment # 03

Installation and Introduction of Packet Tracer

Objective:

- Installation of Packet Tracer.
- Establish a connection by using packet tracer and run commands on it.

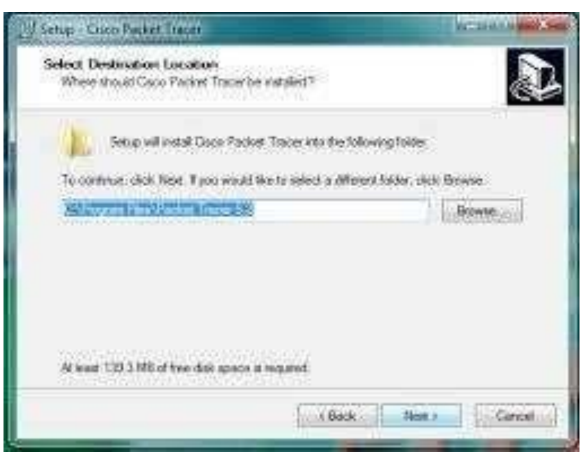
Theory:

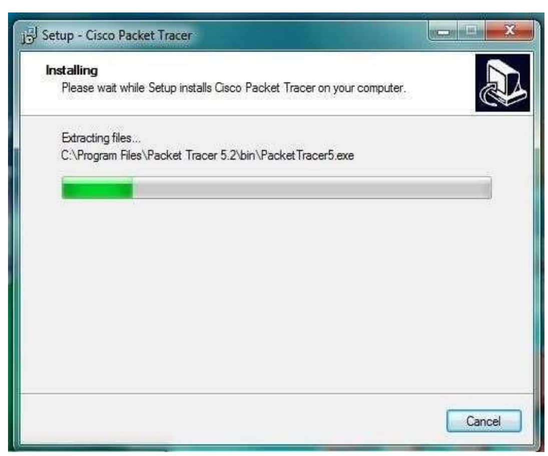
Cisco Packet Tracer is a network simulation program that allows students to experiment with network behavior. Packet Tracer provides simulation, visualization, and authoring, assessment, and collaboration capabilities and facilitates the teaching and learning of complex technology concepts. The current version of Packet Tracer supports an array of simulated Application Layer protocols, as well as basic routing with RIP,OSPF, and EIGRP, to the extents required by the current CCNA curriculum. While Packet Tracer aims to provide a realistic simulation of functional networks, the application itself utilizes only a small number of features found within the actual hardware running a current Cisco IOS version. Thus, Packet Tracer is unsuitable for modeling production networks. It can be used to design a network by connecting various networking devices and running various troubleshooting tests to check the connectivity and communication between different networking devices. Packet Tracer can be used to understand the use of different networking devices appropriately and the difference in their working. As it is costly to buy various networking equipment while learning networking, Packet Tracer can be used to understand computer networks.

Procedure:

1. Installation:

First of all install the software by name packet tracer which we use for establishing connections between two or more devices. I install the software by clicking next, next, and then finish.

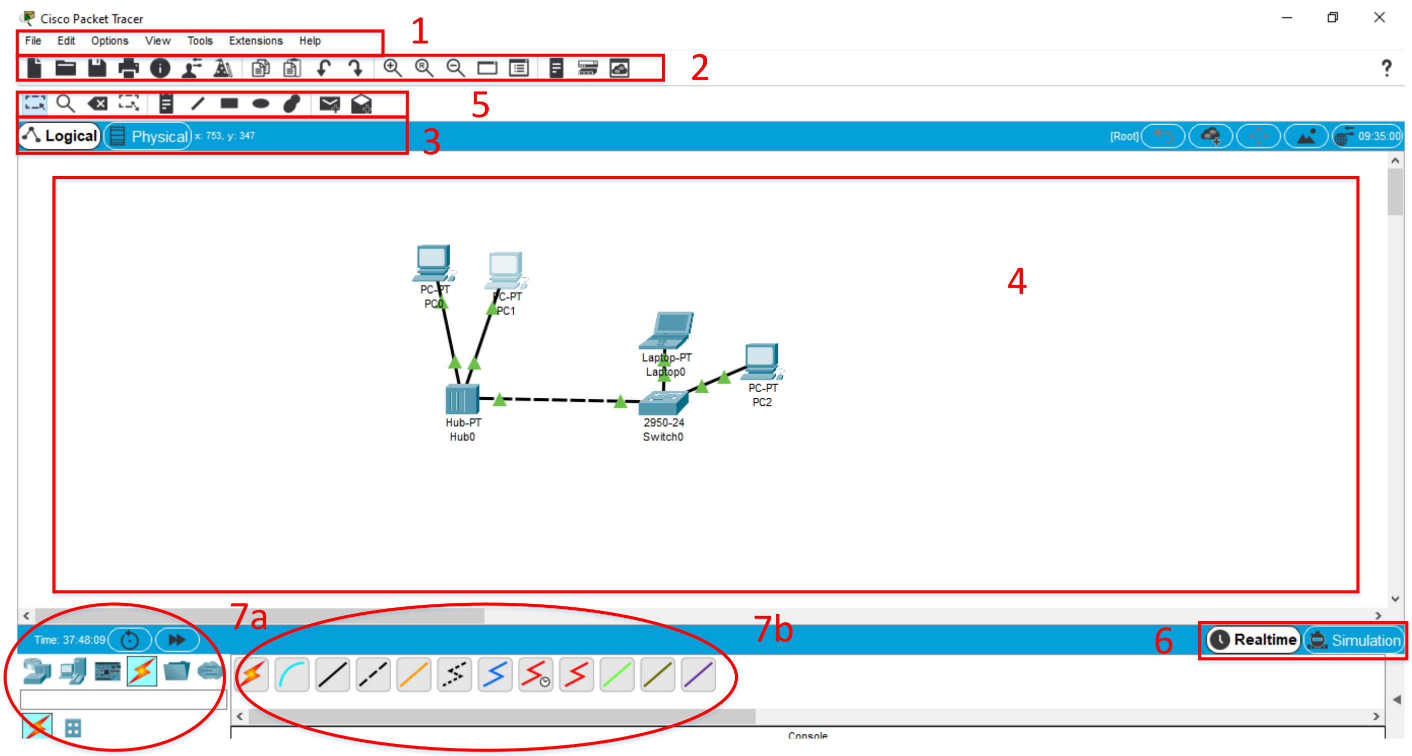




Procedure:

1. Interface overview

The layout of Packet Tracer is divided into several components similar to a photo editor. Match the numbering in the following screenshot with the explanations given after it:



The components of the Packet Tracer interface are as follows:

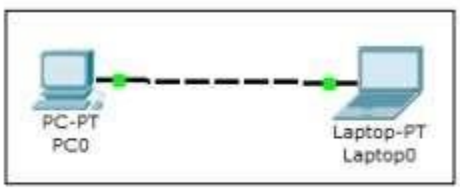
Area 1: Menu bar – This is a common menu found in all software applications; it is used to open, save, print, change preferences, and so on.

Area 2: Main toolbar – This bar provides shortcut icons to menu options that are commonly accessed, such as open, save, zoom, undo, and redo, and on the right-hand side is an icon for entering network information for the current network.
Area 3: Logical/Physical workspace tabs – These tabs allow you to toggle between the Logical and Physical work areas.
Area 4: Workspace – This is the area where topologies are created, and simulations are displayed.
Area 5: Common tools bar – This toolbar provides controls for manipulating topologies, such as select, move layout, place note, delete, inspect, resize shape, and add simple/complex PDU.
Area 6: Realtime/Simulation tabs – These tabs are used to toggle between the real and simulation modes. Buttons are also provided to control the time, and to capture the packets.
Area 7: Network component box – This component contains all the network and end devices available with Packet Tracer, and is further divided into two areas:
° **Area 7a: Device-type selection box** – This area contains device categories
° **Area 7b: Device-specific selection box** – When a device category is selected, this selection box displays

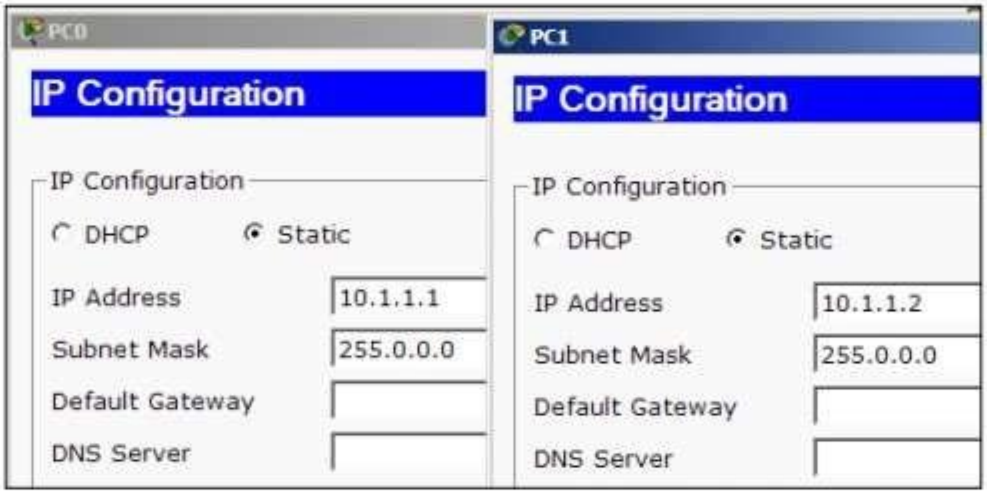
2. Creating a Simple Topology

Now that you're familiar with the GUI of Packet Tracer, you can create your first network topology by carrying out the following steps:

1. From the network component box, click on End Devices and drag-and-drop a Generic PC icon and a Generic laptop icon into the Workspace.



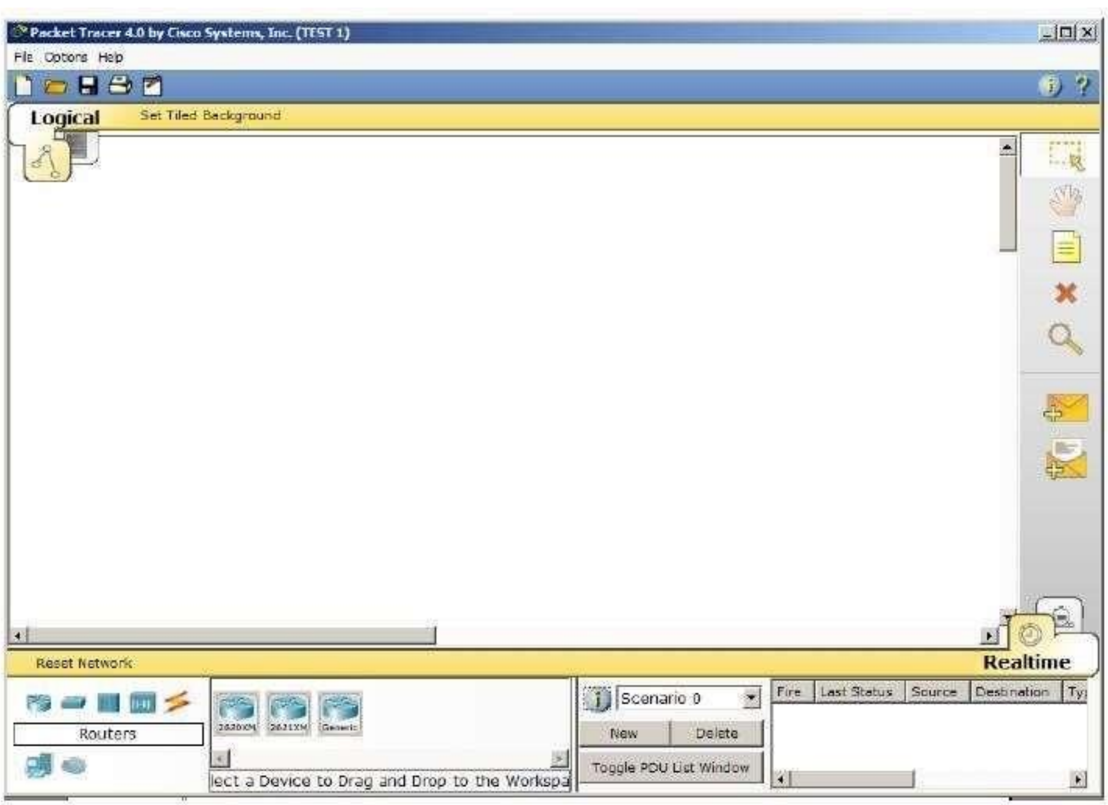
2. Click on Connections, then click on Copper Cross-Over, then on PC0, and select Fast Ethernet. After this, click on Laptop0 and select Fast Ethernet. The link status LED should show up in green, indicating that the link is up.
3. Click on the PC, go to the Desktop tab, click on IP Configuration, and enter an IP address and subnet mask. In this topology, the default gateway and DNS server information is not needed as there are only two end devices in the network.
4. Close the window, open the laptop, and assign an IP address to it in the same way. Make sure that both of the IP addresses are in the same subnet.



Close the IP Configuration box, open the command prompt, and ping the IP address of the device at the end to check connectivity.

3. Introduction to the Packet Tracer Interface using a Hub Topology

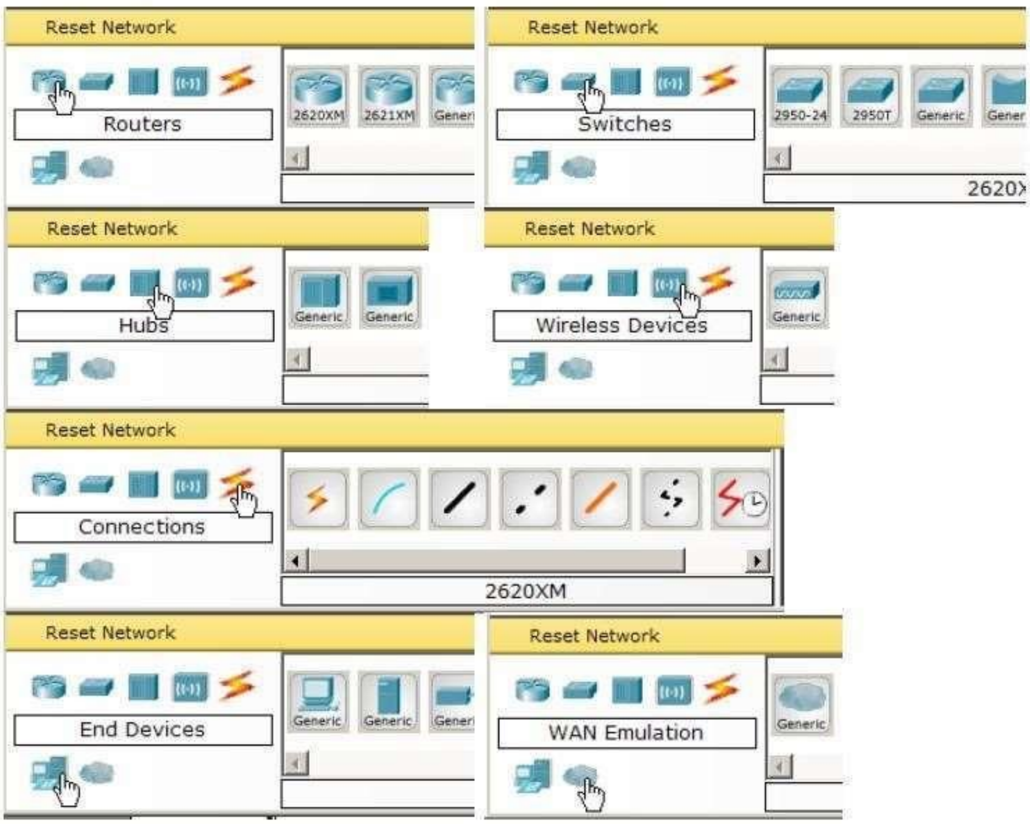
Step 1: Start Packet Tracer and Entering Simulation Mode



Step 2: Choosing Devices and Connections

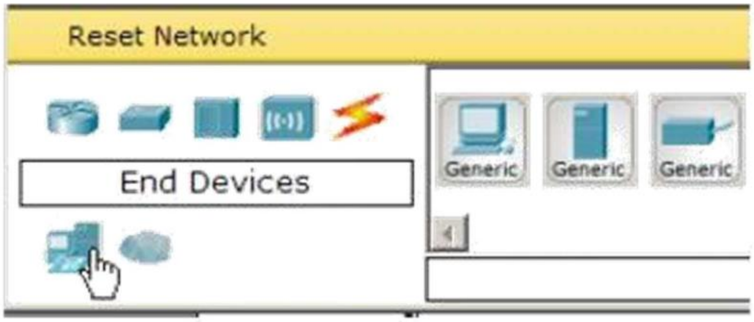
We will begin building our network topology by selecting devices and the media in which to connect them.
Several types of devices and network connections can be used. For this lab we will keep it simple by using End Devices, Switches, Hubs, and Connections.

Single click on each group of devices and connections to display the various choices.

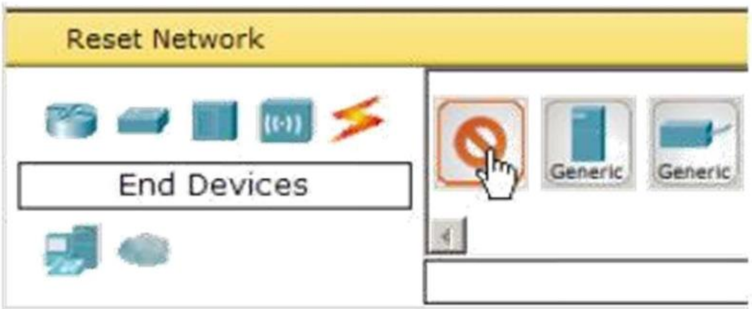


Step 3: Building the Topology – Adding Hosts

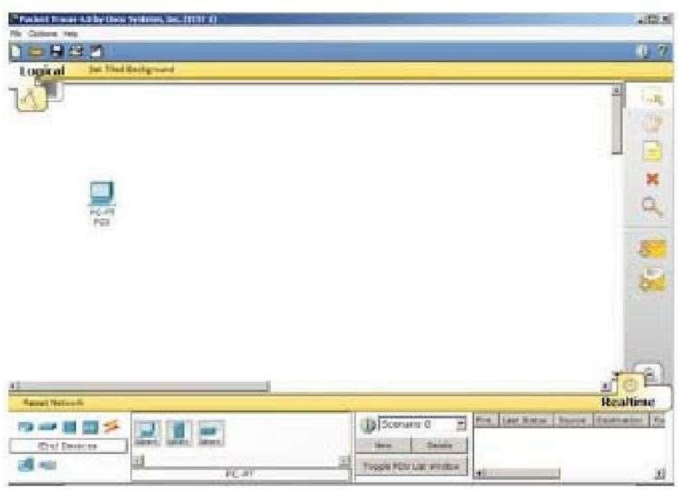
Single click on the End Devices.



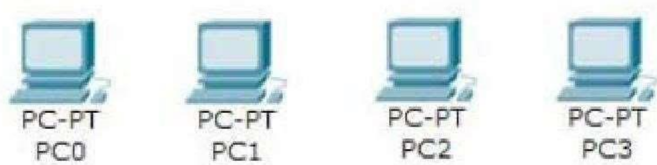
Single click on the **Generic** host.



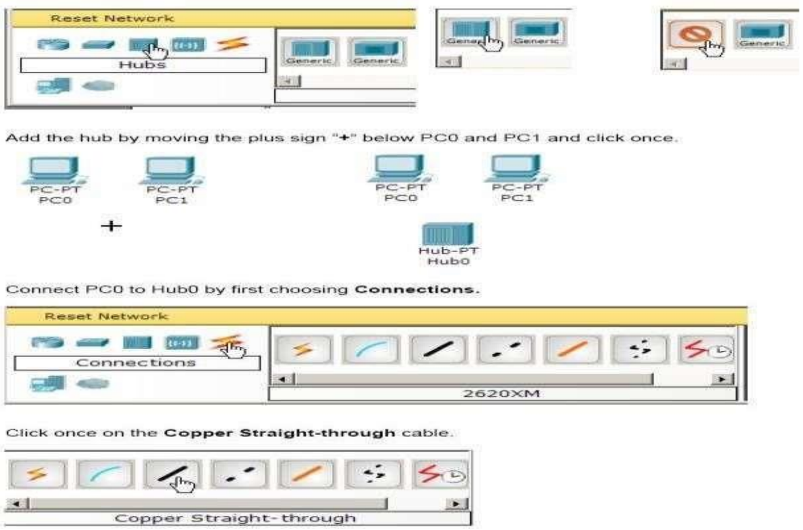
Move the cursor into topology area. You will notice it turns into a plus "+" sign. Single click in the topology area and it copies the device.



Add three more hosts.

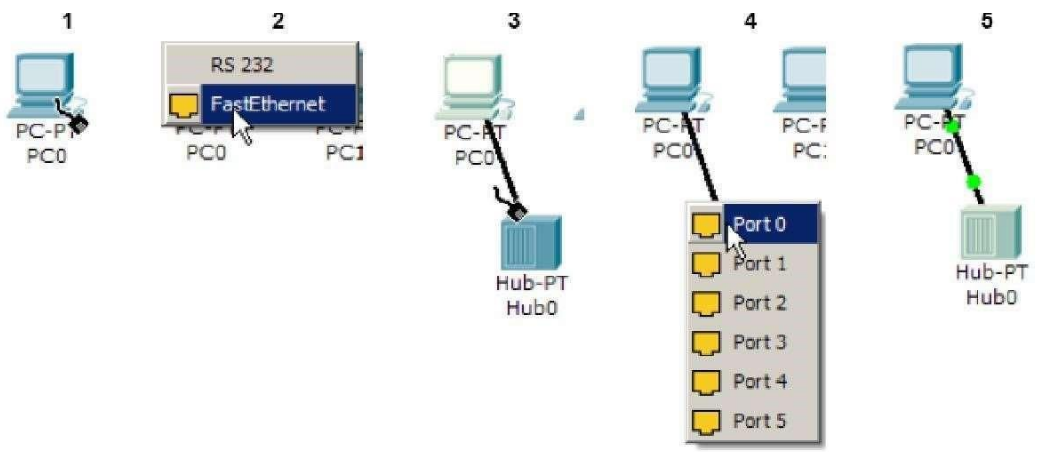


Step 4: Building the Topology – Connecting the Hosts to Hubs and Switches
Adding a Hub
Select a hub, by clicking once on Hubs and once on a Generic hub.

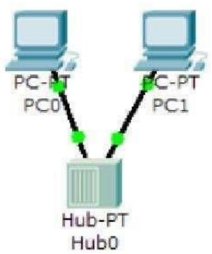


Perform the following steps to connect PC0 to Hub0:

- Click once on PC0
- Choose FastEthernet
- Drag the cursor to Hub0
- Click once on Hub0 and choose Port 0
- Notice the green link lights on both the PC0 Ethernet NIC and the Hub0 Port 0 showing that the link is active.



Repeat the steps above for **PC1** connecting it to **Port 1** on **Hub0**. (The actual hub port you choose does not matter.)



Adding a Switch

Select a switch, by clicking once on Switches and once on a 2950-24 switch.

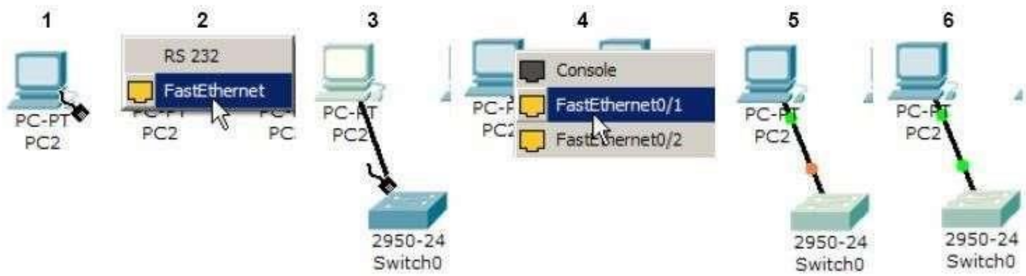
Add the switch by moving the plus sign "+" below PC2 and PC3 and click once.

Connect PC2 to Hub0 by first choosing Connections.

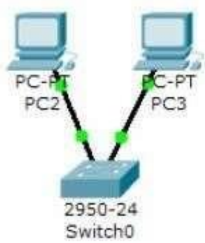
Click once on the Copper Straight-through cable.

Perform the following steps to connect PC2 to Switch0:

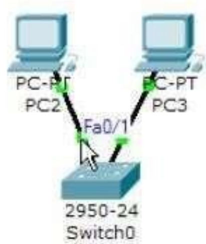
1. Click once on PC2
2. Choose Fast Ethernet
3. Drag the cursor to Switch0
4. Click once on Switch0 and choose FastEthernet0/1
5. Notice the green link lights on PC2 Ethernet NIC and amber light Switch0 FastEthernet0/1 port. The switch port is temporarily not forwarding frames, while it goes through the stages for the Spanning Tree Protocol (STP) process.
6. After a about 30 seconds the amber light will change to green indicating that the port has entered the forwarding stage. Frames can now forward out the switch port.



Repeat the steps above for **PC3** connecting it to **Port 3** on **Switch0** on port **FastEtherent0/2**. (The actual switch port you choose does not matter.)



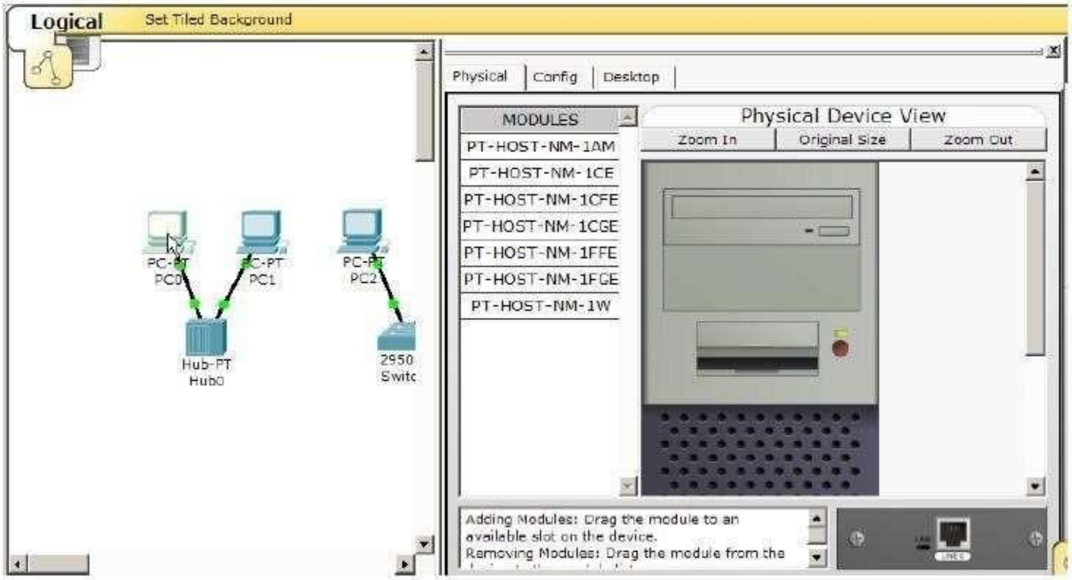
Move the cursor over the link light to view the port number. **Fa** means FastEthernet, 100 Mbps Ethernet.



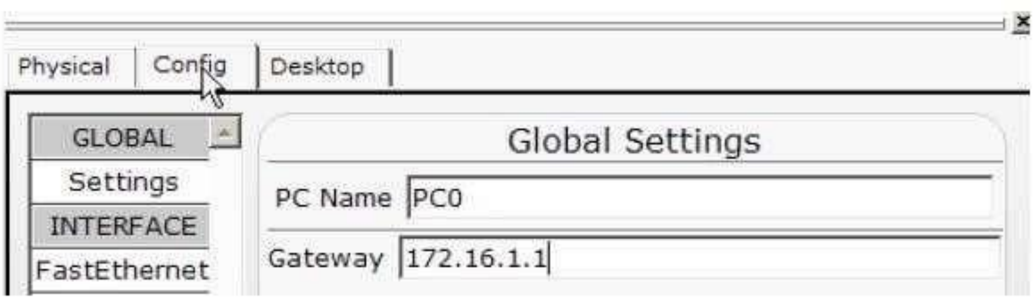
Step 5: Configuring IP Addresses and Subnet Masks on the Hosts

Before we can communicate between the hosts we need to configure IP Addresses and Subnet Masks on the devices.

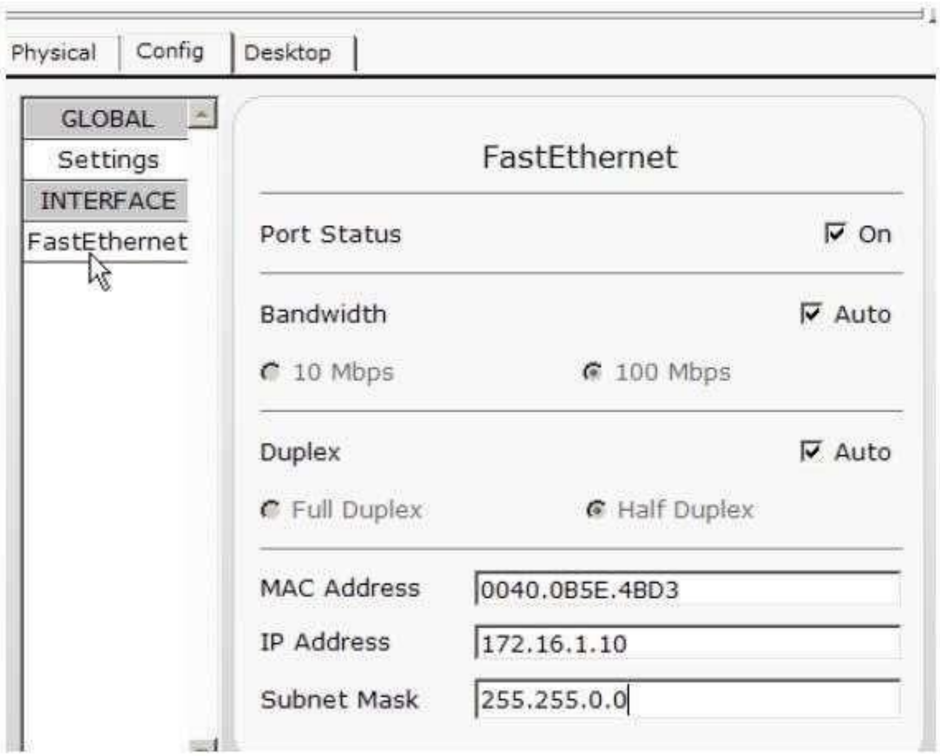
Click once on PC0.



Choose the Config tab. It is here that you can change the name of PC0. Itis also here where you would enter a Gateway IP Address, also known as the default gateway. We will discuss this later, but this would be the IP address of the local router. If you want, you can enter the IP Address 172.16.1.1, although it will not be used in this lab.



Click on Fast Ethernet. Although we have not yet discussed IP Addresses, add the IP Address to 172.16.1.10. Click once in the Subnet Mask field to enter the default Subnet Mask. You can leave this at 255.255.0.0. We will discuss this later.



Also, notice this is where you can change the Bandwidth (speed) and Duplex of the Ethernet NIC (Network Interface Card). The default is Auto (auto negotiation), which means the NIC will negotiate with the hub or switch. The bandwidth and/or duplex can be manually set by removing the check from the Auto box and choosing the specific option.

Bandwidth

- Auto

If the host is connected to a hub or switch port which can do 100 Mbps, then the Ethernet NIC on the host will choose 100 Mbps (Fast Ethernet). Otherwise, if the hub or switch port can only do 10 Mbps, then the Ethernet NIC on the host will choose 10 Mbps (Ethernet).

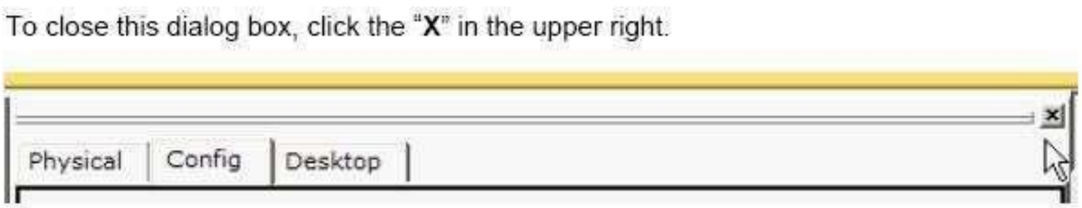
Duplex - Auto

Hub: If the host is connected to a hub, then the Ethernet NIC on the host will choose Half Duplex.

Switch: If the host is connected to a switch, and the switch port is configured as Full Duplex (or Auto negotiation), then the Ethernet NIC on the host will choose Full Duplex.
If the switch port is configured as Half Duplex, then the

Ethernet NIC on the host will choose Half Duplex. (Full Duplex is a much more efficient option.)

The information is automatically saved when entered.



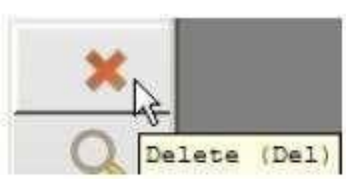
Repeat these steps for the other hosts. Use the information below for IP Addresses and Subnet Masks.

Host	IP Address	Subnet Mask
PC0	172.16.1.10	255.255.0.0
PC1	172.16.1.11	255.255.0.0
PC2	172.16.1.12	255.255.0.0
PC3	172.16.1.13	255.255.0.0

Verify the information
To verify the information that you entered, move the Select tool (arrow) over each host.

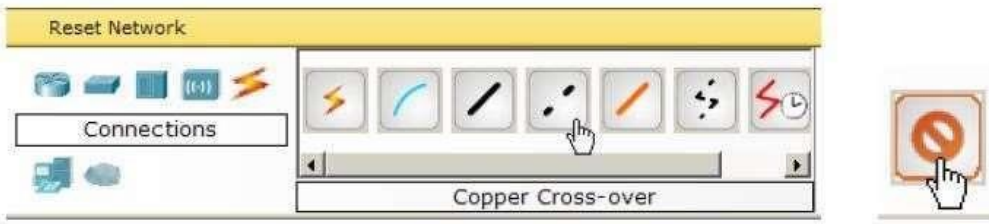
Deleting a Device or Link

To delete a device or link, choose the Delete tool and click on the item you wish to delete.

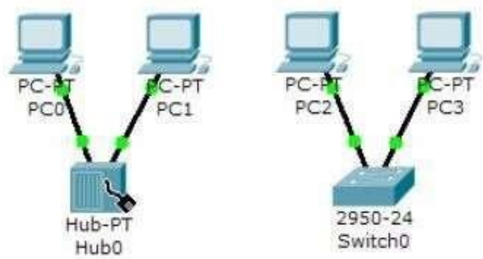


Step 6: Connecting Hub0 to Switch0

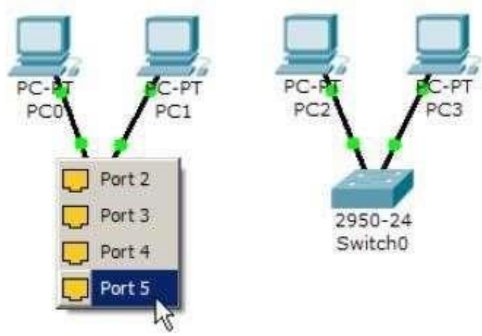
To connect like-devices, like a Hub and a Switch, we will use a Cross-over cable. Click the Cross-over Cable once from the Connections options.



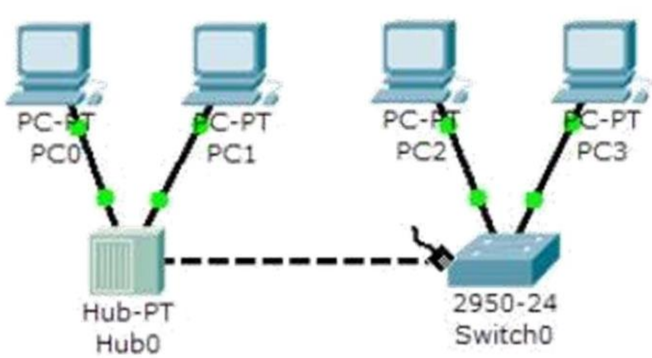
Move the Connections cursor over **Hub0** and click once.



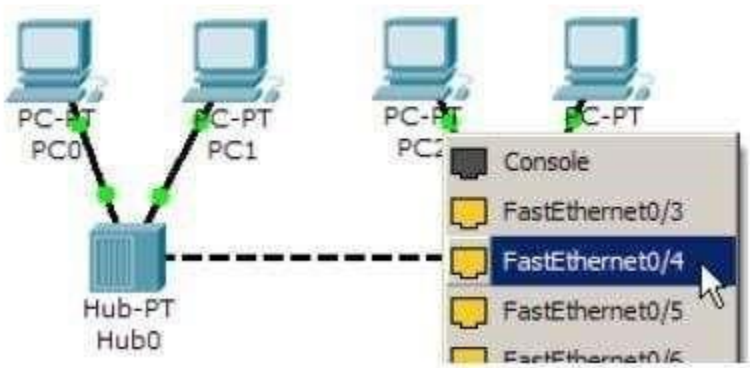
Select **Port 5** (actual port does not matter).



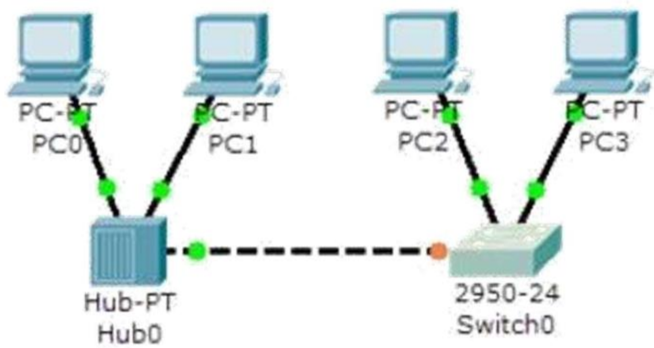
Move the Connections cursor to Switch0.

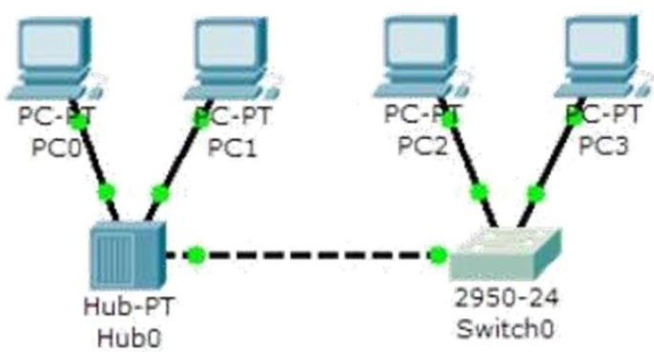


Click once on Switch0 and choose FastEthernet0/4 (actual port does not matter).



The link light for switch port FastEthernet0/4 will begin as amber and eventually change to green as the Spanning Tree Protocol transitions the port to forwarding.





LAB TASKS

1. Simulate above network devices to understand about CISCO Packet Tracer.
2. Explain why the connection fails if the bandwidth or duplex settings for the devices do not match.
3. Explain, If a device is configured with bandwidth and duplex settings on automatic, does the device at the other end of the connection have to be set to automatic as well?

Conclusion